

Qualification Pack



Apiculturist (wild bee) NTFP (Non-Timber Forest Produce)

QP Code: SGJ/Q3301

Version: 1.0

NSQF Level: 2

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SGJ/Q3301: Apiculturist (wild bee) NTFP (Non-Timber Forest Produce)

Brief Job Description

NTFP includes all the Non- Timber Forest Products that are harvested from trees, shrubs and other plants in the forests. The course on (I) Value Addition & Marketing of Non-Timber Forest Products (NTFPs)- (Animal Origin): Wild Beekeeping & processing A Beekeeper is responsible for carrying out beekeeping operations which include preparing for and setting up beehives procuring and introducing bees in hives, and maintaining the bees and beehives. The individual is also responsible for harvesting, processing and marketing honey and related produce.

Personal Attributes

Apiculturist focusing on wild bees and Non-Timber Forest Produce (NTFP) include a passion for nature, environmental stewardship, adaptability, patience, observational skills, collaborative spirit, innovative thinking, respect for indigenous knowledge, effective communication skills, and resilience.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N0205: Introduce Animal based NTFP management](#)
2. [SGJ/N0206: Perform Basics of Colony organization/Colony division, Colony united and Bee handling and management](#)
3. [SGJ/N0207: Identify Methods for Apiary & Grafting, Honey/Wax Extraction, Pollination and other natural requirements](#)
4. [SGJ/N0208: Perform Bee colony capturing from wild and hive and Bee Botany and Preparation of floral calendar](#)
5. [SGJ/N0209: Exposure Visit- opportunity to interact with successful beekeepers](#)
6. [SGJ/N0210: Ensure Management of resources and Opportunities](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Forestry

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Occupation	Forestry and related workers
Country	India
NSQF Level	2
Credits	8
Aligned to NCO/ISCO/ISIC Code	6210.9900(Forestry and Related Workers) 6113.9900 (Nursery Workers and Gardeners, Other) 6115(Medicinal and Aromatic Plant Cultivators) 6210.0600(Gatherer, Medicinal Herbs) 6111.0100(Cultivator, General) 6113.0200 (Nurseryman)
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	Not Applicable
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/01/2027
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-02-ES-01753-2023-V1-SCGJ
NQR Version	1.0

Remarks:

Total 240 hours (Theory: 65 hours, Practical: 145 hours and Employability Skills: 30 hours)

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SGJ/N0205: Introduce Animal based NTFP management

Description

This unit tells about Animal based NTFP management

Scope

The scope covers the following :

- For animal-based NTFP management encompasses ecological understanding, sustainable harvesting techniques, processing methods, and adaptive monitoring to distinguish and achieve sustainable development goals and understanding of indigenous traditional knowledge and its role in NTFP management.
- The history of beekeeping and its economic importance entails examining the evolution of apiculture practices, background of wild beekeeping & processing and its role in livelihood generation.

Elements and Performance Criteria

Animal based NTFP management (Common Module)

To be competent, the user/individual on the job must be able to:

- PC1.** introduction to NTFPs- Resources, potential in economy and livelihood generation.
- PC2.** explain conservation Awareness and approaches.
- PC3.** discuss indigenous traditional knowledge and its role in NTFP management.
- PC4.** discuss sustainable and harvesting: Principle and practices.
- PC5.** discuss intellectual Property Rights and the Patent Regime.
- PC6.** understanding of NTFP Market dynamics.
- PC7.** overview of NTFPs: Resources, Economic Potential, and Generation of Livelihoods; Conservation Awareness and Methods.
- PC8.** demonstrate native American customary knowledge and its function in managing the NTFP
- PC9.** perform harvesting and sustainability: Concept and methods.
- PC10.** illustrate rights to Intellectual Property and the Patent System.
- PC11.** demonstrate knowledge of the dynamics of the NTFP Market.

History of Beekeeping, Its economic importance (Theory)

To be competent, the user/individual on the job must be able to:

- PC12.** explain type of Bee Products (Honey, Royal jelly, pollen and Bee colony)
- PC13.** discuss demand of consumer related to bee products.
- PC14.** explain wild Honey Bee species -Apis cerana Indica,, Apis dorsata, Apis florae and Trigona species.
- PC15.** explain life cycle of Honey Bee.
- PC16.** discuss apis cerana Indica-External Morphology and Internal Morphology, Behaviour and Ecology
- PC17.** explain type of Beekeeping Equipments for Wild honey bee-A. cerana.
- PC18.** Illustrate about Bee Products: pollen, honey, royal jelly, and bee colonies.

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- PC19.** identify consumer demand for products derived from bees.
- PC20.** show the Apis cerana Indica, Apis dorsata, Apis florae, and Trigona species are wild honey bee species.
- PC21.** show the Apis cerana Indica life cycle includes external and internal morphology, behavior, and ecology.
- PC22.** show kind of Equipment Used in Beekeeping for A. cerana, the Wild Honey Bee.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** understanding the ecology of the animals and their habitats.
- KU2.** developing sustainable harvesting practices that minimize impact on populations.
- KU3.** processing and marketing products in a way that adds value and generates income.
- KU4.** ensuring equitable benefit sharing among stakeholders.
- KU5.** promoting responsible ecotourism practices.
- KU6.** generates income for beekeepers, creates jobs in related industries, and contributes to rural livelihoods.
- KU7.** bees are key pollinators, ensuring the reproduction of many plants and contributing to biodiversity.
- KU8.** beekeeping has deep cultural significance in many societies and is associated with traditional knowledge and practices.
- KU9.** continuous learning and adaptation are crucial due to the dynamic nature of both animal-based NTFP management and beekeeping practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** a strong understanding of animal biology, ecology, and conservation principles is essential for sustainably managing animal-based NTFPs and beekeeping practices.
- GS2.** the ability to effectively communicate with stakeholders, including local communities, government agencies, and businesses, is crucial for collaboration and successful project implementation.
- GS3.** the ability to identify different animal species relevant to NTFP management and handle them safely and ethically is essential.
- GS4.** for beekeeping, specific skills in hive management, honey extraction, and disease control are necessary.
- GS5.** understanding market dynamics, pricing strategies, and value addition for animal-based NTFPs and bee products is crucial for economic success.
- GS6.** the ability to plan, implement, and monitor projects related to animal-based NTFP management and beekeeping is essential for achieving desired outcomes.
- GS7.** this involves knowing how to extract honey from hives without harming the bees and how to process it for market.



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- GS8.** this involves knowing how to care for bee colonies, including providing them with food and water, controlling pests and diseases, and preventing swarming.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Animal based NTFP management (Common Module)</i>	24	25	-	-
PC1. introduction to NTFPs- Resources, potential in economy and livelihood generation.	4	-	-	-
PC2. explain conservation Awareness and approaches.	4	-	-	-
PC3. discuss indigenous traditional knowledge and its role in NTFP management.	4	-	-	-
PC4. discuss sustainable and harvesting: Principle and practices.	4	-	-	-
PC5. discuss intellectual Property Rights and the Patent Regime.	4	-	-	-
PC6. understanding of NTFP Market dynamics.	4	-	-	-
PC7. overview of NTFPs: Resources, Economic Potential, and Generation of Livelihoods; Conservation Awareness and Methods.	-	5	-	-
PC8. demonstrate native American customary knowledge and its function in managing the NTFP	-	5	-	-
PC9. perform harvesting and sustainability: Concept and methods.	-	5	-	-
PC10. illustrate rights to Intellectual Property and the Patent System.	-	5	-	-
PC11. demonstrate knowledge of the dynamics of the NTFP Market.	-	5	-	-
<i>History of Beekeeping, Its economic importance (Theory)</i>	26	25	-	-
PC12. explain type of Bee Products (Honey, Royal jelly, pollen and Bee colony)	4	-	-	-
PC13. discuss demand of consumer related to bee products.	4	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. explain wild Honey Bee species -Apis cerana Indica,, Apis dorsata, Apis florae and Trigona species.	4	-	-	-
PC15. explain life cycle of Honey Bee.	4	-	-	-
PC16. discuss apis cerana Indica-External Morphology and Internal Morphology, Behaviour and Ecology	5	-	-	-
PC17. explain type of Beekeeping Equipments for Wild honey bee-A. cerana.	5	-	-	-
PC18. Illustrate about Bee Products: pollen, honey, royal jelly, and bee colonies.	-	5	-	-
PC19. identify consumer demand for products derived from bees.	-	5	-	-
PC20. show the Apis cerana Indica, Apis dorsata, Apis florae, and Trigona species are wild honey bee species.	-	5	-	-
PC21. show the Apis cerana Indica life cycle includes external and internal morphology, behavior, and ecology.	-	5	-	-
PC22. show kind of Equipment Used in Beekeeping for A. cerana, the Wild Honey Bee.	-	5	-	-
NOS Total	50	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0205
NOS Name	Introduce Animal based NTFP management
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Forestry and related workers
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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SGJ/N0206: Perform Basics of Colony organization/Colony division, Colony united and Bee handling and management

Description

This unit tells about Basics of Colony organization/Colony division, Colony united and Bee handling and management.

Scope

The scope covers the following :

- Entails comprehensive training in the principles, methods, and best practices for effectively managing honeybee colonies, encompassing colony assessment, swarm retrieval, colony splitting, and colony integration techniques to optimize hive health and productivity.
- Equipping individuals with the knowledge and skills necessary to safely and effectively handle bees, manage hive activities, and implement seasonal management practices to support colony development, health, and productivity throughout the year.

Elements and Performance Criteria

Colony organization/ Colony capturing/ Colony division, Colony united (Theory and Practical)

To be competent, the user/individual on the job must be able to:

- PC1.** discuss queen, New Queen cell formation, fertilized virgin, Queen mating, drone, etc.
- PC2.** swarm colony and Natural colony.
- PC3.** queen less colony control and apply of queen cell, control of worker laying, etc.
- PC4.** showcase queen, New Queen cell formation, fertilized virgin, Queen mating, drone, etc.
- PC5.** identify swarm colony and Natural colony
- PC6.** showcase queen less colony control and apply of queen cell, control of worker laying, etc.

Bee handling and management/Seasonal management (Theory and Practical).

To be competent, the user/individual on the job must be able to:

- PC7.** explain dearth period
- PC8.** explain growth period
- PC9.** explain honey flow period
- PC10.** discuss seasonal management-(Drone control, swarm control and Queen control, Rainy season, etc)
- PC11.** illustrate dearth period management.
- PC12.** illustrate growth period observation
- PC13.** ensure understanding of Honey flow period
- PC14.** perform seasonal management-(Drone control, swarm control and Queen control, Rainy season, etc)

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** understanding the ecology of the animals and their habitats.
- KU2.** developing sustainable harvesting practices that minimize impact on populations.
- KU3.** processing and marketing products in a way that adds value and generates income.
- KU4.** ensuring equitable benefit sharing among stakeholders.
- KU5.** promoting responsible ecotourism practices.
- KU6.** generates income for beekeepers, creates jobs in related industries, and contributes to rural livelihoods.
- KU7.** bees are key pollinators, ensuring the reproduction of many plants and contributing to biodiversity.
- KU8.** beekeeping has deep cultural significance in many societies and is associated with traditional knowledge and practices.
- KU9.** continuous learning and adaptation are crucial due to the dynamic nature of both animal-based NTFP management and beekeeping practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** a strong understanding of animal biology, ecology, and conservation principles is essential for sustainably managing animal-based NTFPs and beekeeping practices.
- GS2.** the ability to effectively communicate with stakeholders, including local communities, government agencies, and businesses, is crucial for collaboration and successful project implementation.
- GS3.** the ability to identify different animal species relevant to NTFP management and handle them safely and ethically is essential.
- GS4.** for beekeeping, specific skills in hive management, honey extraction, and disease control are necessary.
- GS5.** understanding market dynamics, pricing strategies, and value addition for animal-based NTFPs and bee products is crucial for economic success.
- GS6.** the ability to plan, implement, and monitor projects related to animal-based NTFP management and beekeeping is essential for achieving desired outcomes.
- GS7.** this involves knowing how to extract honey from hives without harming the bees and how to process it for market.
- GS8.** this involves knowing how to care for bee colonies, including providing them with food and water, controlling pests and diseases, and preventing swarming.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Colony organization/ Colony capturing/ Colony division, Colony united (Theory and Practical)</i>	13	10	-	-
PC1. discuss queen, New Queen cell formation, fertilized virgin, Queen mating, drone, etc.	4	-	-	-
PC2. swarm colony and Natural colony.	4	-	-	-
PC3. queen less colony control and apply of queen cell, control of worker laying, etc.	5	-	-	-
PC4. showcase queen, New Queen cell formation, fertilized virgin, Queen mating, drone, etc.	-	2	-	-
PC5. identify swarm colony and Natural colony	-	3	-	-
PC6. showcase queen less colony control and apply of queen cell, control of worker laying, etc.	-	5	-	-
<i>Bee handling and management/Seasonal management (Theory and Practical).</i>	12	15	-	-
PC7. explain dearth period	3	-	-	-
PC8. explain growth period	3	-	-	-
PC9. explain honey flow period	3	-	-	-
PC10. discuss seasonal management-(Drone control, swarm control and Queen control, Rainy season, etc)	3	-	-	-
PC11. illustrate dearth period management.	-	4	-	-
PC12. illustrate growth period observation	-	4	-	-
PC13. ensure understanding of Honey flow period	-	4	-	-
PC14. perform seasonal management-(Drone control, swarm control and Queen control, Rainy season, etc)	-	3	-	-
NOS Total	25	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0206
NOS Name	Perform Basics of Colony organization/Colony division, Colony united and Bee handling and management
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Forestry and related workers
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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SGJ/N0207: Identify Methods for Apiary & Grafting, Honey/Wax Extraction, Pollination and other natural requirements

Description

This unit tells about Methods for Apiary & Grafting, Honey/Wax Extraction, Pollination and other natural requirements

Scope

The scope covers the following :

- Enhancing agricultural productivity by promoting pollination and facilitating the propagation of desirable plant traits, respectively, thereby contributing to sustainable farming practices and improved crop yields.
- Encompasses the sustainable utilization of bee products, fostering economic opportunities for beekeepers while promoting environmental conservation through the production of natural, renewable resources.
- Safeguarding ecosystems, preserving biodiversity, and ensuring the reproductive success of flora and fauna, thereby supporting food security, ecosystem resilience, and understand the honey bee in relation with agriculture and pesticide poisoning, honey production in relation to water.

Elements and Performance Criteria

Apiary & Grafting method

To be competent, the user/individual on the job must be able to:

- PC1.** discuss site selection, bee flora, seasonal flower, water, shade.
- PC2.** explain bee colony migration.
- PC3.** explain type and uses of grafting equipment.
- PC4.** define artificial Queen rearing, etc.
- PC5.** show importance of site selection, Bee flora, seasonal flower, water, shade.
- PC6.** perform bee colony migration and management.
- PC7.** demonstrate type and Uses of grafting equipment.
- PC8.** ensure understanding about Artificial Queen rearing, etc.

Honey/Wax Extraction

To be competent, the user/individual on the job must be able to:

- PC9.** explain uses of honey, Honey bee wax and its uses.
- PC10.** explain quality control of honey, extraction-filtration, indirect heating and storage.
- PC11.** explain wax extraction.
- PC12.** discuss about bee Enemies/ Pest/Diseases and management.
- PC13.** ensure practical understanding about the uses of honey, Honey bee wax and its uses, quality control of honey, extraction-filtration, indirect heating and storage.
- PC14.** ensure practical understanding about Bee Enemies/Pest/Diseases and management.

Pollination and other natural requirements

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To be competent, the user/individual on the job must be able to:

- PC15.** discuss about honey Bee in relation with agriculture.
- PC16.** explain honey bee and pesticide poisoning.
- PC17.** discuss honey production in relation to water.
- PC18.** ensure practical understanding the honey bee in relation with agriculture and pesticide poisoning, honey production in relation to water.
- PC19.** ensure practical understanding the role of organic farming and horticulture in relation with wild honeybee.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** choose a sunny, sheltered location with access to water and diverse foraging sources (flowers, trees).
- KU2.** decide on hive type (Langstroth, Top Bar, etc.) based on your experience and goals.
- KU3.** acquire essential tools like hive tools, smokers, protective suit, frames, and extractors.
- KU4.** regularly check colony health, brood pattern, queen presence, and potential issues like swarming or disease.
- KU5.** raising new queens to replace old ones or create specific breeding stock.
- KU6.** maintaining optimal queen cell development within hives or queen rearing setups.
- KU7.** honeybees are crucial pollinators for crops, fruits, and flowers, ensuring plant reproduction and biodiversity.
- KU8.** avoid using harmful pesticides near apiaries to protect bee health and foraging success.
- KU9.** ensure a clean and accessible water source near the hives for bees' hydration and hive cooling.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** knowledge of bee biology and behavior.
- GS2.** hive inspection and monitoring skills.
- GS3.** disease and pest identification and control methods.
- GS4.** safe and gentle bee handling techniques.
- GS5.** record-keeping and data analysis.
- GS6.** understanding local regulations and best practices.
- GS7.** advanced knowledge of hive management techniques (e.g., swarm control, queen rearing).
- GS8.** understanding queen bee biology and development & basic beekeeping skills and equipment handling.
- GS9.** knowledge of honey extraction equipment and its operation & proper honey hygiene and sanitation practices.
- GS10.** understanding honey processing and storage methods & basic marketing and sales skills.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Apiary & Grafting method</i>	9	8	-	-
PC1. discuss site selection, bee flora, seasonal flower, water, shade.	2	-	-	-
PC2. explain bee colony migration.	2	-	-	-
PC3. explain type and uses of grafting equipment.	2	-	-	-
PC4. define artificial Queen rearing, etc.	3	-	-	-
PC5. show importance of site selection, Bee flora, seasonal flower, water, shade.	-	2	-	-
PC6. perform bee colony migration and management.	-	2	-	-
PC7. demonstrate type and Uses of grafting equipment.	-	2	-	-
PC8. ensure understanding about Artificial Queen rearing, etc.	-	2	-	-
<i>Honey/Wax Extraction</i>	12	4	-	-
PC9. explain uses of honey, Honey bee wax and its uses.	3	-	-	-
PC10. explain quality control of honey, extraction-filtration, indirect heating and storage.	3	-	-	-
PC11. explain wax extraction.	3	-	-	-
PC12. discuss about bee Enemies/ Pest/Diseases and management.	3	-	-	-
PC13. ensure practical understanding about the uses of honey, Honey bee wax and its uses, quality control of honey, extraction-filtration, indirect heating and storage.	-	2	-	-
PC14. ensure practical understanding about Bee Enemies/Pest/Diseases and management.	-	2	-	-
<i>Pollination and other natural requirements</i>	9	8	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. discuss about honey Bee in relation with agriculture.	3	-	-	-
PC16. explain honey bee and pesticide poisoning.	3	-	-	-
PC17. discuss honey production in relation to water.	3	-	-	-
PC18. ensure practical understanding the honey bee in relation with agriculture and pesticide poisoning, honey production in relation to water.	-	4	-	-
PC19. ensure practical understanding the role of organic farming and horticulture in relation with wild honeybee.	-	4	-	-
NOS Total	30	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0207
NOS Name	Identify Methods for Apiary & Grafting, Honey/Wax Extraction, Pollination and other natural requirements
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Forestry and related workers
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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SGJ/N0208: Perform Bee colony capturing from wild and hive and Bee Botany and Preparation of floral calendar

Description

This unit tells how to Perform Bee colony capturing from wild and hive and Bee Botany and Preparation of floral calendar

Scope

The scope covers the following :

- Understand the Bee colony capturing from wild and introduction into hive
- Gain Practical knowledge about Bee Botany and Preparation of floral calendar

Elements and Performance Criteria

Bee colony capturing from wild and introduction into hive

To be competent, the user/individual on the job must be able to:

- PC1.** explain overview of the process why and when bee colonies are captured from the wild.
- PC2.** discuss behavioral characteristics of wild bee colonies.
- PC3.** explain Techniques of Bee Colony Capturing.
- PC4.** explain Hive selection and preparation for introducing wild colonies.
- PC5.** perform Bee colony capturing from wild and introduction into hive

Bee Botany and Preparation of floral calendar

To be competent, the user/individual on the job must be able to:

- PC6.** explain definition and significance of bee botany in apiculture.
- PC7.** discuss and understand the relationship between bees and flowering plants.
- PC8.** explain importance of floral resources for bee nutrition and honey production.
- PC9.** explain Procedure for Herbarium collection and preservation.
- PC10.** perform Practical exercises in documenting flowering periods of selected plant species.
- PC11.** show how to do classification of floral resources: nectar, pollen, and other rewards.
- PC12.** perform Herbarium collection and preservation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** observe bee activity near potential nesting sites like hollow trees, rock crevices, or human-made structures. Scout for honeybee entrance holes or characteristic sounds.
- KU2.** use humane methods like bait hives, swarm traps, or gentle shaking techniques. Prioritize bee welfare and minimize stress.
- KU3.** ensure permits and landowner permissions are obtained before capturing wild colonies.

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- KU4.** purchase from beekeepers or reputable sources. Consider breed suitability, health status, and equipment compatibility.
- KU5.** ensure proper ventilation, temperature control, and minimal disturbance during transport.
- KU6.** learn to identify common nectar and pollen-producing plants in your region. Focus on flowering seasons and their overlap to create a continuous food source.
- KU7.** choose diverse, bee-friendly plants with staggered blooming times throughout the season. Consider local ecology and avoid pesticide-treated areas.
- KU8.** design a bee-friendly garden or landscape with a variety of flowering plants, shrubs, and trees. Provide water sources and nesting sites.
- KU9.** research and record flowering times of major nectar and pollen sources in your area.
- KU10.** track blooming periods throughout the season and adjust beekeeping practices accordingly. This helps identify potential gaps in food availability and plan supplementary feeding if needed.
- KU11.** a floral calendar helps ensure colony health and productivity by anticipating potential food shortages and taking proactive measures.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** identifying potential swarms, understanding bee behavior, and waiting for the right moment to capture are crucial.
- GS2.** bees are naturally defensive, so using protective gear and handling them with respect is essential.
- GS3.** be aware of local laws and regulations regarding capturing wild bees and hiving them.
- GS4.** knowing how to use swarm boxes, capture nets, and other tools safely and effectively.
- GS5.** capturing swarms can be challenging, and experience helps anticipate their behavior and handle unexpected situations.
- GS6.** building relationships with local beekeepers can provide valuable knowledge and support.
- GS7.** identifying blooming plants that provide pollen and nectar for bees is important for choosing hive locations and supporting local ecosystems.
- GS8.** knowing when different plants bloom helps predict honey flow and manage hive health.
- GS9.** understanding how plants attract pollinators and the factors that influence nectar and pollen production.
- GS10.** learn about the specific flowering patterns and pollen/nectar quality of native and cultivated plants in your area.
- GS11.** understanding the relationship between bees and specific plants allows for targeted management practices.
- GS12.** promoting diverse landscapes and bee-friendly plants fosters healthy bee populations and benefits the environment.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Bee colony capturing from wild and introduction into hive</i>	8	20	-	-
PC1. explain overview of the process why and when bee colonies are captured from the wild.	2	15	-	-
PC2. discuss behavioral characteristics of wild bee colonies.	2	-	-	-
PC3. explain Techniques of Bee Colony Capturing.	2	-	-	-
PC4. explain Hive selection and preparation for introducing wild colonies.	2	-	-	-
PC5. perform Bee colony capturing from wild and introduction into hive	-	5	-	-
<i>Bee Botany and Preparation of floral calendar</i>	7	15	-	-
PC6. explain definition and significance of bee botany in apiculture.	1	-	-	-
PC7. discuss and understand the relationship between bees and flowering plants.	2	-	-	-
PC8. explain importance of floral resources for bee nutrition and honey production.	2	-	-	-
PC9. explain Procedure for Herbarium collection and preservation.	2	-	-	-
PC10. perform Practical exercises in documenting flowering periods of selected plant species.	-	5	-	-
PC11. show how to do classification of floral resources: nectar, pollen, and other rewards.	-	5	-	-
PC12. perform Herbarium collection and preservation	-	5	-	-
NOS Total	15	35	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0208
NOS Name	Perform Bee colony capturing from wild and hive and Bee Botany and Preparation of floral calendar
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Forestry and related workers
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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SGJ/N0209: Exposure Visit- opportunity to interact with successful beekeepers

Description

This unit tells about how to explore the Exposure Visit- opportunity to interact with successful beekeepers

Scope

The scope covers the following :

- Interact with successful beekeepers lies in providing valuable learning experiences, practical insights, and networking opportunities to aspiring beekeepers, facilitating knowledge exchange, skill development, and the promotion of best practices in beekeeping.

Elements and Performance Criteria

Exposure Visit- opportunity to interact with successful beekeepers

To be competent, the user/individual on the job must be able to:

- PC1.** discuss setting objectives and expectations for the visit
- PC2.** discuss identification of target beekeepers and scheduling visits.
- PC3.** discuss Purpose of exposure visits: knowledge exchange, skill development, and inspiration.
- PC4.** do exposure Visit- opportunity to interact with successful beekeepers and demonstration by the master trainer

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** understand the different roles of worker bees, drones, and queens, their lifecycle stages, and how they contribute to the hive.
- KU2.** familiarize yourself with the different parts of a beehive, including the brood chamber, honey supers, frames, and queen excluder. Learn about the various tools and equipment used by beekeepers.
- KU3.** develop basic understanding of bee communication, foraging strategies, swarming, and defense mechanisms.
- KU4.** research various beekeeping methods such as Langstroth, Top Bar, and Warre hives, along with their pros and cons.
- KU5.** learn about hive inspections, brood management, swarm control, disease and pest identification, and honey extraction techniques.
- KU6.** understand the importance of sustainable beekeeping practices, including integrated pest management, habitat preservation, and pollinator-friendly planting.
- KU7.** gain insights into the financial aspects of beekeeping, including production costs, marketing strategies, and pricing considerations.

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- KU8.** explore different ways beekeepers sell their honey and other bee products, such as through farmers markets, subscriptions, or creating value-added products like lip balm or candles.
- KU9.** be aware of any local regulations or legal requirements related to beekeeping in your area.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** what aspects of beekeeping are you most interested in learning about? Focus your research on those areas.
- GS2.** understand that successful beekeepers have likely faced challenges and setbacks along the way. Don't expect to have all the answers by the end of your visit.
- GS3.** come prepared with thoughtful questions for the beekeepers you will meet. This demonstrates your enthusiasm and allows you to delve deeper into specific topics.
- GS4.** learn about their specific practices, successes, and challenges beforehand.
- GS5.** wear long pants, closed-toe shoes, and protective clothing as recommended.
- GS6.** pay attention, ask thoughtful questions, and avoid interrupting.
- GS7.** document key information and capture visuals for future reference.
- GS8.** follow up with the beekeepers after your visit and maintain communication.
- GS9.** thank the beekeepers for their time and insights.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Exposure Visit- opportunity to interact with successful beekeepers</i>	20	30	-	-
PC1. discuss setting objectives and expectations for the visit	5	-	-	-
PC2. discuss identification of target beekeepers and scheduling visits.	5	-	-	-
PC3. discuss Purpose of exposure visits: knowledge exchange, skill development, and inspiration.	10	-	-	-
PC4. do exposure Visit- opportunity to interact with successful beekeepers and demonstration by the master trainer	-	30	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0209
NOS Name	Exposure Visit- opportunity to interact with successful beekeepers
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Forestry and related workers
NSQF Level	2
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N0210: Ensure Management of resources and Opportunities

Description

This unit tells about how to Ensure Management of resources and Opportunities

Scope

The scope covers the following :

- Enterprise management, men, material and money Knowledge on supporting agencies-KVIC, NABARD, DICCC, SIDBI, MSME, Banks, etc.

Elements and Performance Criteria

Management of resources and Opportunities

To be competent, the user/individual on the job must be able to:

- PC1.** discuss knowledge about Enterprise management, men, material and money.
- PC2.** discuss knowledge on supporting agencies-KVIC, NABARD, DICCC, SIDBI, MSME, Banks, etc.
- PC3.** show knowledge about Enterprise management, men, material and money
- PC4.** show knowledge on supporting agencies-KVIC, NABARD, DICCC, SIDBI, MSME, Banks, etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** knowledge of hygienic and efficient methods for extracting, filtering, and storing honey to maintain its quality and safety is essential.
- KU2.** exploring and developing new products like beeswax candles, propolis extracts, or pollen-based supplements can increase income and diversify offerings.
- KU3.** understanding market trends, consumer preferences, and effective marketing strategies is crucial for successfully selling bee products.
- KU4.** tracking costs, income, and resource use helps make informed decisions and ensure the economic viability of the operation.
- KU5.** implementing practices that minimize environmental impact, such as reducing waste and supporting local ecosystems, is crucial for long-term success.
- KU6.** sharing knowledge about bees and sustainable practices with local communities can foster support and collaboration.
- KU7.** connecting with experienced beekeepers and joining relevant organizations can provide valuable knowledge, resources, and networking opportunities.
- KU8.** universities, research centers, and government agencies often offer workshops, publications, and extension services related to beekeeping and sustainable practices.
- KU9.** knowledge of how to create and maintain suitable habitat for wild bees, including providing diverse flowering plants and nesting sites, is crucial for their long-term survival.

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- KU10.** knowing the different species present in your area, their specific needs, and their preferred forage sources is essential for managing colonies effectively and supporting populations.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** ability to create and implement a business plan that considers market needs, financial viability, and sustainable practices.
- GS2.** skills in promoting and selling bee products to diverse audiences, including understanding market trends and customer preferences.
- GS3.** ability to manage finances effectively, including budgeting, pricing, and cost control.
- GS4.** ability to maintain accurate financial records and report on income, expenses, and profits.
- GS5.** understanding and mitigating potential risks associated with wild beekeeping, such as weather events, disease, and market fluctuations.
- GS6.** building relationships with other beekeepers to share knowledge, resources, and opportunities.
- GS7.** ability to identify and address challenges associated with wild beekeeping, such as resource limitations, equipment malfunctions, and pest infestations.
- GS8.** ability to communicate effectively with diverse stakeholders, including beekeepers, landowners, customers, and regulatory agencies.
- GS9.** ability to stay informed about new research and best practices, and adapt your approach based on changing conditions and market demands.
- GS10.** understanding and implementing practices that ensure the long-term health and viability of wild bee populations and ecosystems.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Management of resources and Opportunities</i>	20	30	-	-
PC1. discuss knowledge about Enterprise management, men, material and money.	10	-	-	-
PC2. discuss knowledge on supporting agencies-KVIC, NABARD, DICCC, SIDBI, MSME, Banks, etc.	10	-	-	-
PC3. show knowledge about Enterprise management, men, material and money	-	15	-	-
PC4. show knowledge on supporting agencies-KVIC, NABARD, DICCC, SIDBI, MSME, Banks, etc.	-	15	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0210
NOS Name	Ensure Management of resources and Opportunities
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Forestry and related workers
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council/Awarding Body. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC/AB will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC/AB.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0205.Introduce Animal based NTFP management	50	50	0	0	100	16
SGJ/N0206.Perform Basics of Colony organization/Colony division, Colony united and Bee handling and management	25	25	0	0	50	14
SGJ/N0207.Identify Methods for Apiary & Grafting, Honey/Wax Extraction, Pollination and other natural requirements	30	20	0	0	50	14
SGJ/N0208.Perform Bee colony capturing from wild and hive and Bee Botany and Preparation of floral calendar	15	35	0	0	50	14
SGJ/N0209.Exposure Visit-opportunity to interact with successful beekeepers	20	30	0	0	50	14
SGJ/N0210.Ensure Management of resources and Opportunities	20	30	0	0	50	14
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	14
Total	180	220	0	0	400	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.